The saturated rainbow cell complex resolves precisely the rainbow ideals with linear resolution

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Editorial note. Exact corrected published version only.

Editorial note. Original author text and native diagrams retained; only counter and wrapper presentation additions.

Editorial note. Unchanged macros, official bibliography and license evidence reused by explicit paths from ALCO500.

Editorial note. r002 restores unchanged native section 5 macros from official published.tex lines 1879–1889; r001 failed on missing susst. No author mathematical text changed.

Editorial scope. Faithful author material collection only. Root owns assembly, compile, visual QA, SQLite, upload and identity refresh.

Allowed standard prerequisites (editorial).

- Labelled cellular complexes and multigraded homology. Miller–Sturmfels, Combinatorial commutative algebra, published reference [39], Section 4 (MiSt). Standard construction and degree-strand identification, explicitly stated in full as Fact 1.
- Minimal multigraded free resolutions, comparison-map lifting and homology exact sequences. Standard homological algebra over the polynomial ring k[A]. Lifting identity to a homogeneous chain map and the exact homology sequence; the degree-controlled comparison and cokernel induction are current author core and retained in full.

Source and counting boundary

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Only whole Theorem 5.5 (`thm:cell-linres`) is counted. Its full current support, particularly Proposition 5.4, is retained but not separately counted. This is independent of the ball/constructibility Theorem 4.3 collected separately: neither proof uses the other. Remark 5.6 and subsequent corollaries are omitted.

Version boundary. The earlier arXiv:2212.09528v2 was withdrawn on 5 August 2024 because of the authors' stated error in old Lemma 6.17, 1a, Case iii, affecting old Theorem 6.7. The v3 announcement on 13 January 2025 states the error was fixed. Only the actual corrected 2026 official published text is used here. No withdrawn bitight-equivalence argument, version mixing, mathematical repair or certification is supplied. Notices: `https://arxiv.org/abs/2212.09528v2`, `https://arxiv.org/abs/2212.09528v3`.

Presentation. Native author mathematical text and macros are preserved. An independent article wrapper and explicit counter settings restore published numbering. The square figure is retained as native TikZ and the comparison diagram as native xymatrix; no external image is required. The complete official bibliography is reused with original numbers and citation keys. Publisher classes, logos and third-party styles are excluded.

Difficulty and scope. The H2 core is the saturation argument giving the homology gap $p+n+2$, followed by degree-controlled comparison-map and cokernel induction. Miller–Sturmfels [39], Section 4, is standard external background; the current author core is reproduced fully. Saved in-flight identity information ends on 14 October 2026 at 17:36:45 UTC; later overlap remains unconfirmed. This is author-material preparation, not mathematical certification or final global-unique admission.

Let $A_1, A_2, \ldots, A_n$ be disjoint finite sets, and $A = \sqcup_{i=1}^n A_i$ their union. The complete intersection ideal

$$C = (m(A_1), m(A_2), \ldots, m(A_n)) \subseteq k[A]$$

is the SR-ideal of the join of the boundaries of simplices

$$\partial\Delta(A_\bullet) = \partial\Delta(A_1) * \partial\Delta(A_2) * \cdots * \partial\Delta(A_n), \tag{4}$$

3.2 Rainbow ideals

Denote $\mathbf{A} = A_1 \times A_2 \times \cdots \times A_n$. For $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ in $\mathbf{A}$, denote by $[\mathbf{a}]$ the set $\{a_1, a_2, \ldots, a_n\}$.

For $\mathbf{a} \in \mathbf{A}$ let $m(\mathbf{a})$ be the monomial $\prod_{i=1}^n a_i$. Thinking of A_i as a color class, we call such a monomial a *rainbow monomial*. The Alexander dual ideal B of the complete intersection $C = (m(A_1), \ldots, m(A_n))$ is generated by the complements of the facets of $\partial\Delta(A_\bullet)$, and so it is the monomial ideal generated by all the rainbow monomials. Letting (A_i) be the "color ideal" generated by the elements of A_i (considered as variables), then B is the product of these color ideals:

$$B = (m(\mathbf{a}) \,|\, \mathbf{a} \in \mathbf{A}) = \prod_{i=1}^n (A_i).$$

Definition 3.2. For a subset $T \subseteq \mathbf{A}$, denote by $I(T)$ the ideal generated by $m(\mathbf{a})$ for $\mathbf{a} \in T$. Such an ideal is called a *rainbow ideal*. For the complement set $T^c = \mathbf{A} \backslash T$, we usually write $L(T)$ for $I(T^c)$, the *complement* rainbow ideal.

A finite set A gives rise to the geometric simplex spanned by the elements of A:

$$\Delta(A) = \left\{ \sum_{a \in A} \gamma_a \cdot a \,|\, 0 \le \gamma_a \le 1, \sum_{a \in A} \gamma_a = 1 \right\} \subseteq \mathbb{R}^A.$$

Given finite non-empty sets $A_1, A_2, \ldots, A_n$, we get a polytope:

$$\Pi(A_\bullet) = \Delta(A_1) \times \Delta(A_2) \times \cdots \times \Delta(A_n) \subseteq \times_{i=1}^n \mathbb{R}^{A_i} = \mathbb{R}^A.$$

Its dual (or polar) polytope is the join

$$\partial\Delta(A_\bullet) = \partial\Delta(A_1) * \partial\Delta(A_2) * \cdots * \partial\Delta(A_n).$$

The set of vertices of $\Pi(A_\bullet)$ is the set $A_1 \times \cdots \times A_n$.

Notation 5.1. When $R_i \subseteq A_i$ for $i = 1, \ldots, n$ are *non-empty* subsets, we write $R_\bullet \subseteq^* A_\bullet$.

Definition 5.2. The polytope $\Pi(A_\bullet)$ gives rise to a cell complex $C(A_\bullet)$, actually a polyhedral cell complex, whose cells are the $\Pi(R_\bullet)$ with $R_\bullet \subseteq^* A_\bullet$. The dimension of such a cell is $\sum_i |R_i| - n$. A subset $T \subseteq A_1 \times A_2 \times \cdots \times A_n$ induces a cell subcomplex $C(T)$ of $C(A_\bullet)$ consisting of all $\Pi(R_\bullet)$ such $R_1 \times R_2 \times \cdots \times R_n \subseteq T$. We say $C(T)$ is the *saturated* cell sub-complex of $C(A_\bullet)$ associated to T.

The (augmented) chain complex:

$$\tilde{C}_\bullet(T) : C_{-1}(T) \leftarrow C_0(T) \leftarrow C_1(T) \leftarrow \cdots \leftarrow C_p(T) \leftarrow \cdots, \tag{7}$$

where $C_p(T)$ is the k-vector space with basis all cells of $C(T)$ of dimension p (and $C_{-1}(T)$ is the one-dimensional vector space with basis the empty cell), and standard differentials (see (8) below). The homology groups are the reduced homology groups. Note that $\tilde{H}_{-1}(\tilde{C}_\bullet(T)) \neq 0$ iff T is empty.

Let S be the polynomial ring $k[A]$ whose variables are the elements of $A = A_1 \cup \cdots \cup A_n$. A vertex $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ of $\Pi(A_\bullet)$ may be labelled by the monomial $m(\mathbf{a}) = a_1 a_2 \cdots a_n$. The cell $\Pi(R_\bullet)$ is then labelled by the least common multiple of the monomials on its vertices. This is simply

$$m(R_\bullet) = m(R_1) m(R_2) \cdots m(R_n), \text{ where } m(R_i) = \prod_{r \in R_i} r,$$

which is a monomial of degree $\dim \Pi(R_\bullet) + n$.

The (augmented) free cellular complex: For $T \subseteq A_1 \times \cdots \times A_n$, this induces a labelling by monomials of the cell sub-complex $C(T)$. By [39, Section 4] it then gives a *free cellular complex*

$$\tilde{\mathcal{C}}_\bullet(T) : \mathcal{C}_{-1}(T) \xleftarrow{d_0} \mathcal{C}_0(T) \xleftarrow{d_1} \cdots \leftarrow \mathcal{C}_p(T) \leftarrow \cdots,$$

where $\mathcal{C}_p(T)$ is the free S-modules with generators all cells $\Pi(R_\bullet)$ in $C(T)$ of dimension p (and $\mathcal{C}_{-1}(T) = S$). The image of d_0 is $I(T)$. We may put a total order on each A_i. Together with the ordered sequence $A_1, A_2, \ldots, A_n$, this gives a total order on the union $A = \cup_1^n A_i$. Then $R_\bullet \subseteq^* A_\bullet$ inherits a total order on $R = \cup_1^n R_i$. For $b \in R$ denote by $\mathrm{pos}(b)$ its position in this total order. For $p \geq 1$ the differential d^p is then given by

$$\Pi(R_\bullet) \mapsto \sum_{\substack{i=1,\ldots,n \\ b \in R_i, |R_i| \geq 2}} (-1)^{\mathrm{pos}(b)} b \cdot \Pi(R_1, \ldots, R_i \backslash \{b\}, \ldots, R_n). \tag{8}$$

Example 5.3. Let $A_1 = \{a_1, b_1\}$ and $A_2 = \{a_2, b_2\}$ and $T = A_1 \times A_2$. The product polytope $\Pi(A_\bullet)$ is the square labelled as in Figure 6.

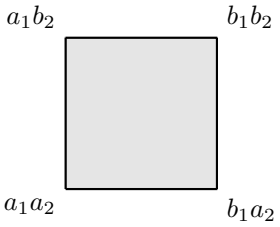


Figure 6: The square $\Pi(A_\bullet)$, with monomial labels on vertices

The augmented chain complex is

$$k \leftarrow k^4 \leftarrow k^4 \leftarrow k,$$

which has no homology, it is acyclic. Let $S = k[a_1, b_1, a_2, b_2]$. The rainbow ideal is $I(T) = (a_1a_2, a_1b_2, b_1a_2, b_1b_2)$. The augmented free cellular complex is

$$S \xleftarrow{d} S(-2)^4 \leftarrow S(-3)^4 \leftarrow S(-4).$$

The image of d is $I(T)$, and in this case, omitting the first term, the complex is a minimal free resolution of $I(T)$. In general, if T is the full $A_1 \times \cdots \times A_n$, the free cellular complex is always a minimal free resolution of $I(T)$. Finding out for which T this free cellular complex *is* a resolution, Theorem 5.5, is an aspect of the main result of our article.

The generators of $\mathcal{C}_p(T)$ for $p \geq 0$ have degrees $p + n$, so $\tilde{\mathcal{C}}_{\geq 0}(T)$ is a *linear* free cellular complex. The free cellular complex $\tilde{\mathcal{C}}_\bullet(T)$ is $\mathbb{N}^A$-graded. A family of subsets $R_\bullet \subseteq^* A_\bullet$ gives a restriction $T_{|R} = T \cap (R_1 \times \cdots \times R_n)$. It also gives a squarefree degree $\deg(R_\bullet) \in \mathbb{N}^A$. The free cellular complex $\tilde{\mathcal{C}}_\bullet(T)_{|\deg(R_\bullet)}$ in this specific degree identifies as the augmented chain complex $\tilde{C}_\bullet(T_{|R_\bullet})$ (see (7)).

Fact 1. *Let $R_\bullet \subseteq^* A_\bullet$. The homology of the augmented free cellular complex $\tilde{\mathcal{C}}_\bullet(T)$ in degree $\deg(R_\bullet) \in \mathbb{N}^A$ is the reduced homology*

$$H_i(\tilde{\mathcal{C}}_\bullet(T))_{\deg(R_\bullet)} = H_i(\tilde{C}_\bullet(T_{|R_\bullet})),\ i \geq -1.$$

This is standard and rather immediate from the above, see [39, Section 4].

Proposition 5.4. *For $p \geq 0$, the homology group $H_p(\tilde{\mathcal{C}}_\bullet(T))$ can only be nonzero in degrees $\geq p + n + 2$.*

Proof. In short, this is a consequence of $C(T)$ being a saturated cell complex. All generators of $\mathcal{C}_p(T)$ are in degree $p + n$. Thus $\ker d_p$ is generated in degrees $\geq p + n + 1$. Let u be a generator of $\ker d_p$ of degree $p + n + 1$. The free cellular complex is $\mathbb{N}^A$-graded, and so we may assume u is homogeneous for

this grading. This degree is also squarefree as it is a minimal generator. Its degree is then $\deg(R_\bullet)$ for some $R_\bullet \subseteq^* A_\bullet$. Consider the free cellular complex $\tilde{\mathcal{C}}_\bullet(R_\bullet)$. We have subcomplexes

$$\tilde{\mathcal{C}}_\bullet(T_{|R_\bullet}) \subseteq \tilde{\mathcal{C}}_\bullet(R_\bullet), \quad \tilde{\mathcal{C}}_\bullet(T_{|R_\bullet}) \subseteq \tilde{\mathcal{C}}_\bullet(T).$$

The latter is an isomorphism in degree $\deg(R_\bullet)$. Thus if $\ker d_p$ has a generator u in degree $\deg(R_\bullet)$, due to $\ker d_p^{\tilde{\mathcal{C}}_\bullet(T_{|R})} \hookrightarrow \ker d_p^{\tilde{\mathcal{C}}_\bullet(R_\bullet)}$ we get a generator of the latter kernel (in degree $\deg(R_\bullet)$), of total degree $p+1+n$. But $\ker d_p^{\tilde{\mathcal{C}}_\bullet(R_\bullet)} = \operatorname{im} d_{p+1}^{\tilde{\mathcal{C}}_\bullet(R_\bullet)}$ and $\mathcal{C}_{p+1}(R_\bullet)$ is the free rank one S-module $S\cdot\Pi(R_\bullet)$, with the generator $\Pi(R_\bullet)$ of degree $p+1+n$. This maps to a sum involving every face of $\Pi(R_\bullet)$ of dimension p. But due to the commutativity of the inclusion maps and differentials of $\tilde{\mathcal{C}}_\bullet(T_{|R_\bullet}) \hookrightarrow \tilde{\mathcal{C}}_\bullet(R_\bullet)$, the degree $p+n$ part $\mathcal{C}_p(T_{|R_\bullet})$ must have each of the generators corresponding to a face of $\Pi(R_\bullet)$ of dimension p. In particular all vertices of $\Pi(R_\bullet)$ must be in T. Since $C(T)$ is saturated, the full cell $\Pi(R_\bullet)$ is in $C(T)$. Then the generator u of $\ker d_p$ is the image of a cell in $\mathcal{C}_{p+1}(T)$. □

Theorem 5.5. *Let $T \subseteq A_1 \times \cdots \times A_n$. If $I(T)$ has linear resolution, the free cellular complex $\tilde{\mathcal{C}}_{\geq 0}(T)$ is its minimal free resolution. (And $\tilde{\mathcal{C}}_{\geq 0}(T)$ is a minimal free resolution precisely in this case.)*

Proof. Let $\mathcal{F}_\bullet$ be a minimal free resolution of $I(T)$. As $I(T)$ is $\mathbb{N}^A$-graded, the same holds for $\mathcal{F}_\bullet$. Since $\mathcal{F}_\bullet$ is a resolution, and $\mathcal{C}_\bullet(T)$ consists of free modules, there is a morphism $\phi : \mathcal{C}_\bullet(T) \to \mathcal{F}_\bullet$ lifting the identity map on $I(T)$. Both $\mathcal{F}_0$ and $\mathcal{C}_0(T)$ have free bases mapping bijectively to the minimal generators of $I(T)$, the monomials $m(\mathbf{a})$ for $\mathbf{a} \in T$. So there is an isomorphism $\mathcal{F}_0 \cong \mathcal{C}_0(T)$.

Assume ϕ_i is an isomorphism for $i \leq p$. Consider the diagram:

$$\begin{array}{ccc} \mathcal{C}_p(T) & \xleftarrow[d^{\mathcal{C}}_{p+1}]{} & \mathcal{C}_{p+1}(T) \\ \downarrow{\scriptstyle \phi_p} & & \downarrow{\scriptstyle \phi_{p+1}} \\ \mathcal{F}_p & \xleftarrow[d^{\mathcal{F}}_{p+1}]{} & \mathcal{F}_{p+1}. \end{array}$$

Since ϕ_p is injective, and $d^{\mathcal{C}}_{p+1}$ is injective on the vector space of generators, which has degree $p+n+1$, the right vertical map ϕ_{p+1} is also injective on the vector space of generators. The map ϕ_{p+1} is then injective since both its domain an codomain are free with generators in degree $p+1+n$.

Let Q_{p+1} be the cokernel of the right vertical map. It is an S-module generated in degree $p+n+1$. By the long exact cohomology sequence of the short exact sequences of complexes

$$0 \to \mathcal{C}_{\leq p+1}(T) \to \mathcal{F}_{\leq p+1} \to Q_{p+1}[-p-1] \to 0$$

(where $[-p-1]$ denotes that Q_{p+1} is in homological degree $p+1$) we see that Q_{p+1} maps surjectively to the homology $H_p(\mathcal{C}_\bullet(T))$. So the latter module is generated in degree $p+n+1$. But it has no homology in this degree, by Proposition 5.4. Hence this homology module vanishes. Then

$$H_{p+1}(\mathcal{F}_{\leq p+1}) \to H_{p+1}(Q_{p+1}[-p-1]) = Q_{p+1}$$

is surjective, where the left module is in fact the kernel $\ker d^{\mathcal{F}}_{p+1}$. Since this kernel is generated in degree $p+n+2$, this again implies that $Q_{p+1} = 0$. Whence ϕ_{p+1} is an isomorphism. □

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